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Studierichting: Informatica
Oefeningenreeks 3G nr. 11.8

$$f(x) = \frac{x-1}{(x+1)^2}$$

1. Domein

$$\text{Dom} f = \mathbb{R} \setminus \{-1\}$$

2. Asymptoten

Verticale asymptoot:

$$x+1=0$$

$$x=-1$$

$$\lim_{x \rightarrow -1^-} \frac{x-1}{(x+1)^2} = -\infty$$

$$\lim_{x \rightarrow -1^+} \frac{x-1}{(x+1)^2} = -\infty$$

$$\Rightarrow \text{VA: } x = -1$$

Horizontale asymptoot:

$$\lim_{x \rightarrow +\infty} \frac{x-1}{(x+1)^2} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{x-1}{(x+1)^2} = 0$$

$$\Rightarrow \text{HA: } y = 0$$

3. Tekenonderzoek van $f(x)$

Teller nul:

$$x-1=0$$

$$x=1$$

$$\Rightarrow f^{(-1)}(0) = \{1\}$$

x	$-\infty$	-1	1	$+\infty$
$f(x)$	$-$	$ $	0	$+$

4. Tekenonderzoek van $f'(x)$

$$f(x) = \frac{x-1}{(x+1)^2}$$

$$f'(x) = \frac{(x+1)^2 \cdot 1 - (x-1) \cdot 2(x+1)}{(x+1)^4}$$

$$f'(x) = \frac{(x+1)[(x+1) - 2(x-1)]}{(x+1)^4}$$

$$f'(x) = \frac{(x+1)(3-x)}{(x+1)^4}$$

$$f'(x) = \frac{3-x}{(x+1)^3}$$

Teller nul:

$$3-x=0$$

$$x=3$$

$$\Rightarrow f'^{(-1)}(0) = \{3\}$$

x	$-\infty$	-1	3	$+\infty$
$f'(x)$	$-$	$ $	$+$	0

Maximum:

$$f(3) = \frac{3-1}{(3+1)^2}$$

$$f(3) = \frac{2}{16}$$

$$f(3) = \frac{1}{8}$$

$$\Rightarrow M\left(3, \frac{1}{8}\right)$$

5. Tekenonderzoek van $f''(x)$

$$f'(x) = \frac{3-x}{(x+1)^3}$$

$$f''(x) = \frac{-1 \cdot (x+1)^3 - (3-x) \cdot 3(x+1)^2}{(x+1)^6}$$

$$f''(x) = \frac{(x+1)^2[-(x+1) - 3(3-x)]}{(x+1)^6}$$

$$f''(x) = \frac{2x-10}{(x+1)^4}$$

Teller nul:

$$2x - 10 = 0$$

$$x = 5$$

$$\Rightarrow f''^{(-1)}(0) = \{5\}$$

x	$-\infty$	-1	5	$+\infty$
$f''(x)$	$-$	$ $	$-$	0
				$+$

Buigpunt:

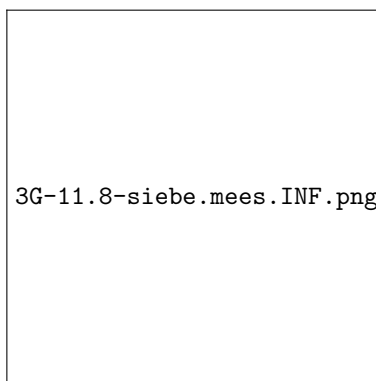
$$f(5) = \frac{5-1}{(5+1)^2}$$

$$f(5) = \frac{4}{36}$$

$$f(5) = \frac{1}{9}$$

$$\Rightarrow B\left(5, \frac{1}{9}\right)$$

6. Schets



$\Rightarrow$ Samenvattende tabel

x	$-\infty$	-1	1	3	5	$+\infty$
$f(x)$	$-$		0	$+$	$+$	$+$
$f'(x)$	$-$		$+$	0	$-$	$-$
$f''(x)$	$-$		$-$	$-$	0	$+$
$f(x)$	$\searrow$		$\nearrow$	$M\left(\frac{1}{8}\right)$	$\searrow$	$\nearrow$
	$\cap$		$\cap$		$B\left(\frac{1}{9}\right)$	$\cup$

References